Lab Assignment 4: Frame-wise Analysis of Speech Signal

Title: To perform frame-wise analysis of a speech signal and study the behaviour of time-domain features for voiced and unvoiced speech.

1. Introduction

Short-time processing is a fundamental technique in speech analysis because speech is a non-stationary signal whose properties change over time. By segmenting the signal into small, quasi-stationary frames (typically 20-30ms), we can extract time-domain features that help distinguish between different types of speech sounds, specifically voiced and unvoiced segments.

2. Methodology & Experimental Setup

2.1 Dataset

- **Sampling Frequency:** 16,000 Hz (16 kHz).
- **Segment Analyzed:** First 20 seconds of the speech sample.

2.2 Framing Parameters

Parameter	Value
Frame Length	25 ms (400 samples)
Frame Shift	10 ms (160 samples)
Total Frames	Calculated based on signal duration

2.3 Window Functions

Two window functions were utilized for feature extraction to compare their effects:

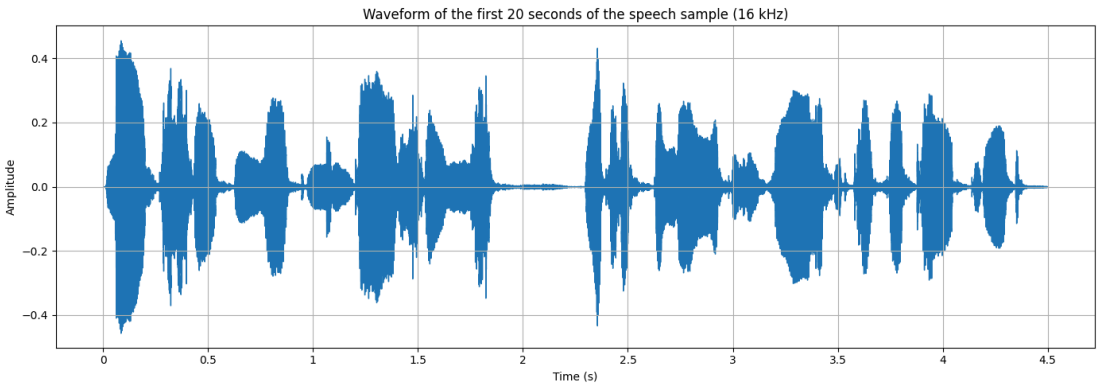
1. **Hamming Window:** Designed to minimize the maximum side-lobe level.
2. **Rectangular Window:** Provides the narrowest main-lobe but higher side-lobes.

3. Task 1: Short-Time Feature Extraction

3.1 Time-Domain Features Extracted

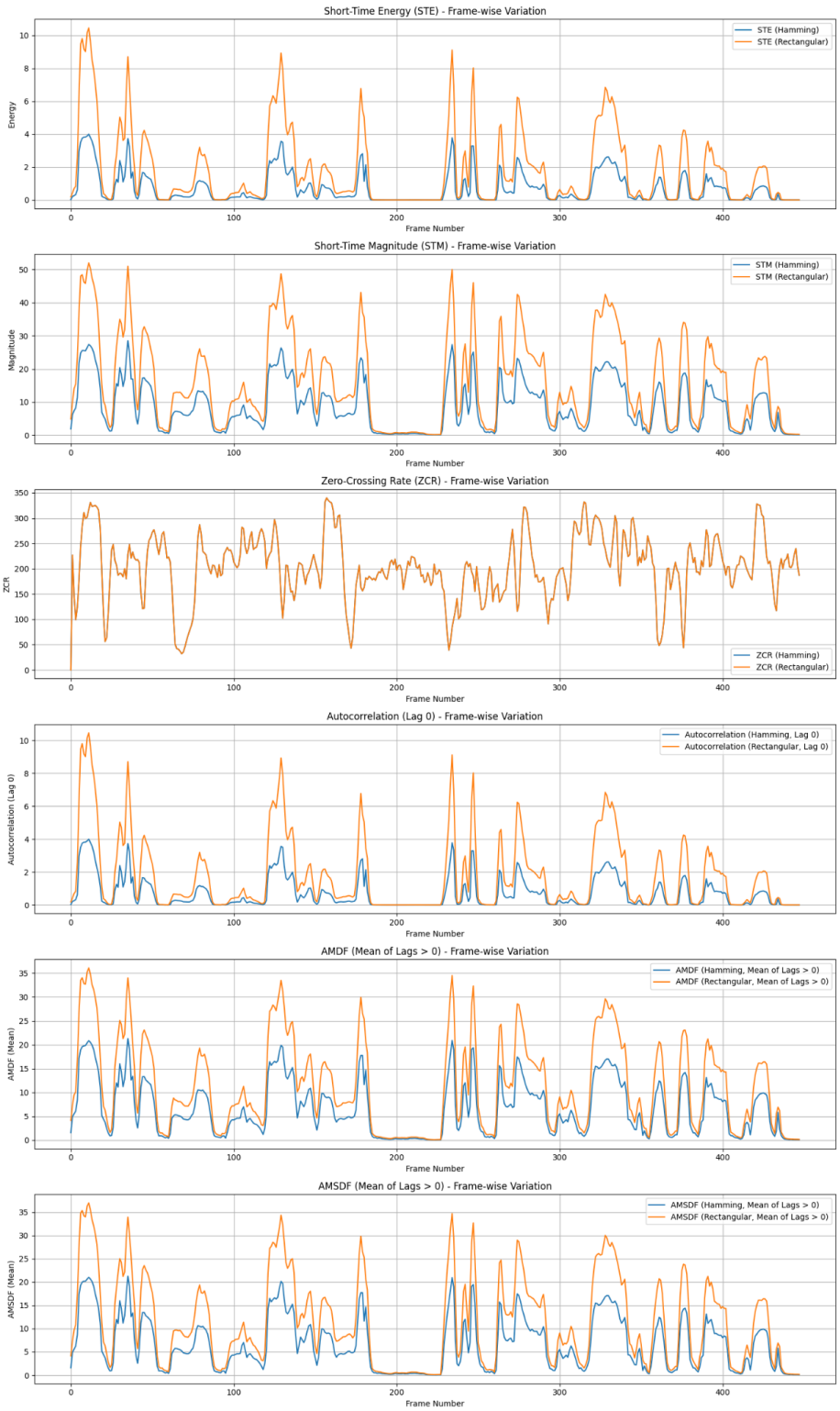
- **Short-Time Energy (STE):** Measures the signal intensity.
- **Short-Time Magnitude (STM):** Similar to energy but less sensitive to high-amplitude peaks.
- **Zero-Crossing Rate (ZCR):** Measures the frequency of sign changes; a proxy for spectral content.
- **Autocorrelation:** Measures self-similarity at different lags.
- **AMDF (Average Magnitude Difference Function):** A difference-based periodicity measure.
- **AMSDF (Average Magnitude Squared Difference Function):** Similar to AMDF but uses squared differences.

3.2 Visual Analysis of Waveform



Description: The waveform shows distinct regions of high-amplitude periodic oscillations (voiced) and low-amplitude noise-like bursts (unvoiced).

3.3 Feature Comparison: Hamming vs. Rectangular

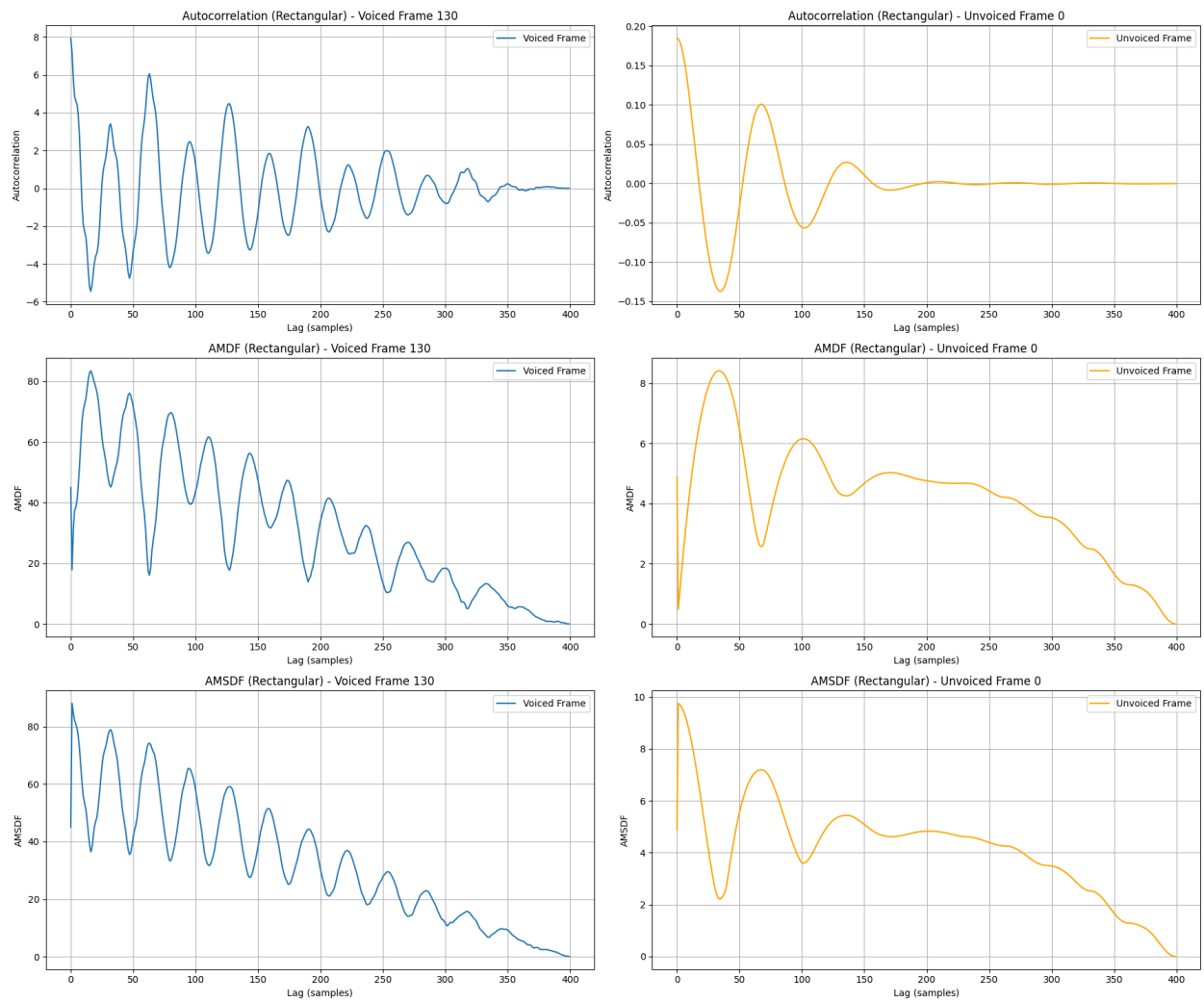


Observation on Feature Smoothness:

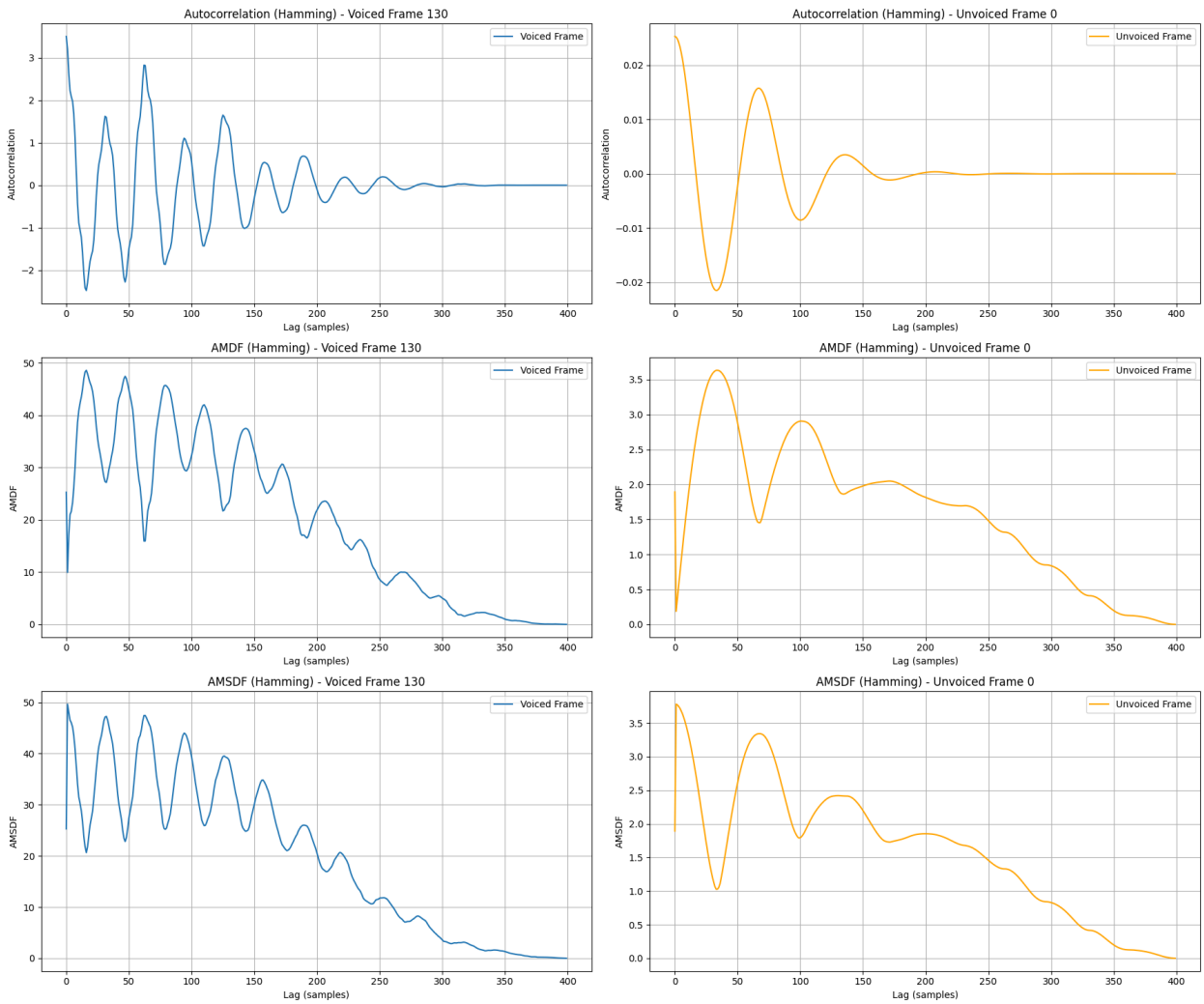
- **Hamming Window:** Produced smoother feature contours. By tapering the edges of each frame, it reduces the impact of discontinuities at frame boundaries, leading to more stable energy and periodicity estimates.
- **Rectangular Window:** Showed more "jitter" or abrupt changes in feature values between adjacent frames. While it preserves higher temporal resolution, the lack of tapering causes spectral leakage and boundary effects.

4. Task 2 & 3: Periodicity Analysis and Pitch Estimation

4.1 Periodic (Voiced) vs. Aperiodic (Unvoiced)



Comparison of periodicity in case of voiced and unvoiced speech signals in Rectangular window



Comparison of periodicity in case of voiced and unvoiced speech signals in Hamming window

Based on the analysis of exemplar frames, the following behaviors were observed across the three periodicity functions:

1. Autocorrelation (ACF)

- Reflection of Periodicity: For voiced frames, the ACF exhibits strong, clear peaks at regular intervals. The first prominent peak after lag 0 indicates the fundamental pitch period. Higher peaks relative to lag 0 signify stronger periodicity.
- Voiced vs. Unvoiced: Voiced frames show these distinct, regularly spaced peaks due to vocal cord vibration. Unvoiced frames exhibit a rapid decay after lag 0 and remain relatively flat or noisy, indicating an absence of repetitive patterns. **The Hamming window results in smoother, better-defined peaks compared to the Rectangular window.**

2. Average Magnitude Difference Function (AMDF)

- Reflection of Periodicity: For voiced frames, the AMDF shows deep, regularly spaced minima (valleys). The deepest minimum (excluding lag 0) corresponds to the fundamental period. **The depth of these valleys indicates the perfection of the signal's repetition.**
- Voiced vs. Unvoiced: Voiced frames display characteristic deep valleys. Unvoiced frames lack these pronounced minima; the curve remains higher and fluctuates irregularly. **The Hamming window smooths these curves, making valleys more discernible.**

3. Average Magnitude Squared Difference Function (AMSDF)

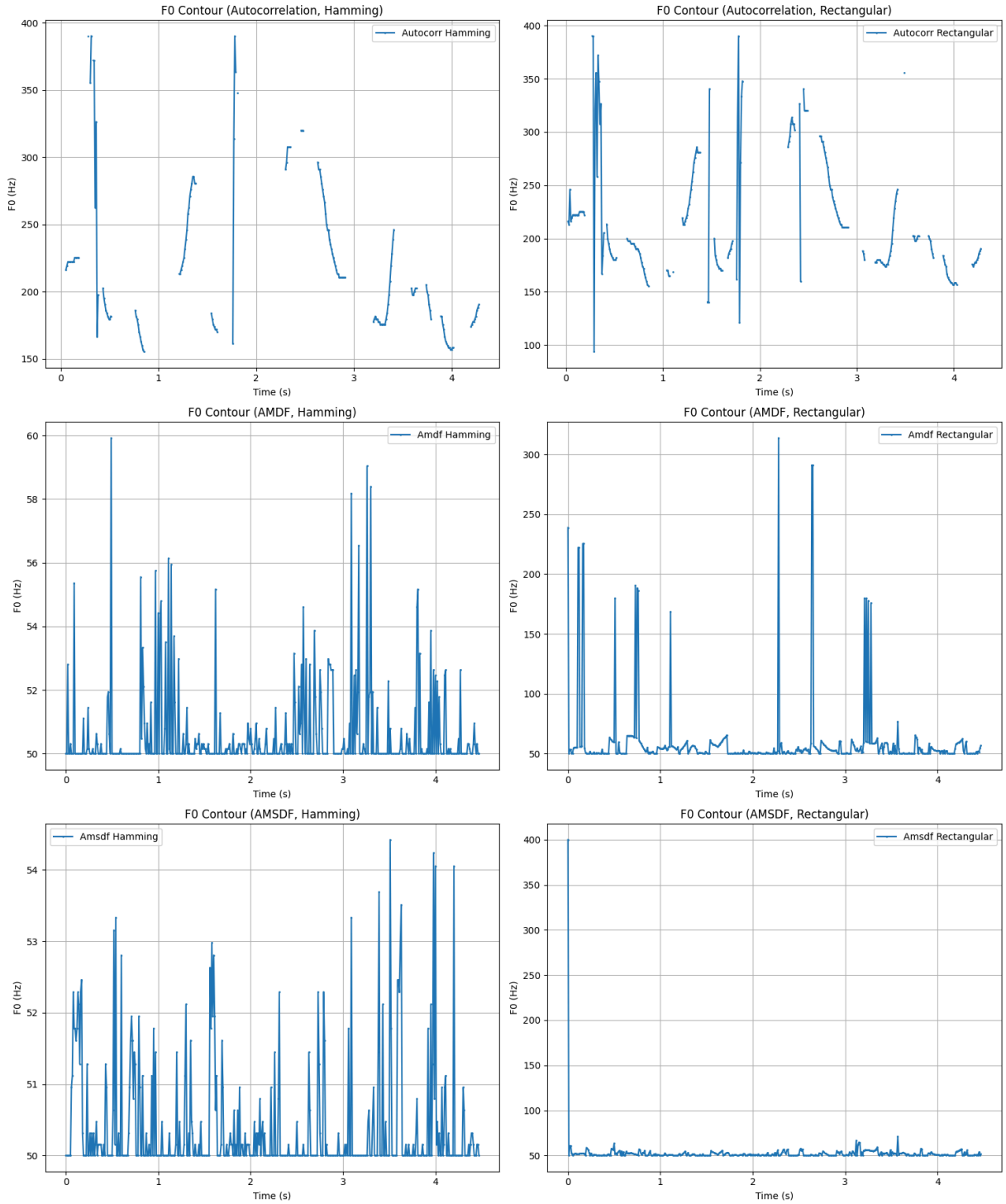
- Reflection of Periodicity: Similar to AMDF, voiced frames exhibit deep, regularly spaced minima where the signal most resembles its shifted version. The lag at the deepest minimum indicates the pitch period.
- Voiced vs. Unvoiced: Voiced frames are characterized by distinct, repetitive minima. Unvoiced frames have relatively high AMSDF values without clear periodic minima. **Again, the Hamming window provides a smoother representation of these periodic traits.**

Summary of Periodicity Indicators

The presence of strong, regularly spaced peaks in Autocorrelation or deep, regularly spaced minima in AMDF/AMSDF are key indicators of a voiced (periodic) speech segment. Conversely, the absence of such clear, regular patterns, often accompanied by a more chaotic or rapidly decaying/fluctuating curve, signifies an unvoiced (aperiodic) speech segment. The choice of window function primarily affects the smoothness and absolute magnitudes of these feature curves, but the fundamental patterns indicating periodicity remain consistent.

4.2 Pitch Frequency (F0) Results

Based on the periodicity analysis, the fundamental frequency was estimated by finding the lag (k) corresponding to the first prominent peak (Autocorr) or minimum (AMDF).



For pitch estimation, **AMDF and AMSDF generally provided more stable and complete F0 contours across voiced segments compared to Autocorrelation, which sometimes showed octave errors or sparse results**. The Hamming window consistently produced smoother and more reliable pitch contours across all methods, effectively reducing noise and artifacts due to its tapering effect. In contrast, the Rectangular window often resulted in slightly noisier F0 estimates. Therefore, the combination of AMDF/AMSDF with a Hamming window appears to be the most effective for robust and smooth pitch estimation in this context.

5. Conclusion

Feature Effectiveness (Task 1 & 2): Short-Time Energy (STE), Short-Time Magnitude (STM), Zero-Crossing Rate (ZCR), Autocorrelation, Average Magnitude Difference Function (AMDF), and Average Magnitude Sum Difference Function (AMSDF) proved effective in differentiating speech characteristics. Voiced segments consistently exhibited higher energy/magnitude features and lower ZCR, while unvoiced segments showed the opposite trend. Autocorrelation peaks and AMDF/AMSDF minima clearly indicated periodicity in voiced speech.

Impact of Windowing (Task 1, 2 & 3): The Hamming window consistently provided smoother and more stable feature representations, which translated to more reliable periodicity detection and F0 estimation. Its tapering effect effectively reduced spectral leakage and edge artifacts. In contrast, the Rectangular window, while preserving absolute magnitudes, often led to noisier feature curves and less smooth F0 contours. ZCR, being less sensitive to amplitude envelopes, remained largely unaffected by the window choice.

Pitch Estimation (Task 3): For F0 estimation, AMDF and AMSDF emerged as more robust and reliable methods compared to Autocorrelation. They generally produced more continuous and stable pitch contours, particularly when applied with a Hamming window, effectively minimizing octave errors and providing a clearer representation of the speech's periodicity. Autocorrelation, while capable, was more prone to inconsistencies and sparser F0 tracks.

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